



AI Energy Control Tower

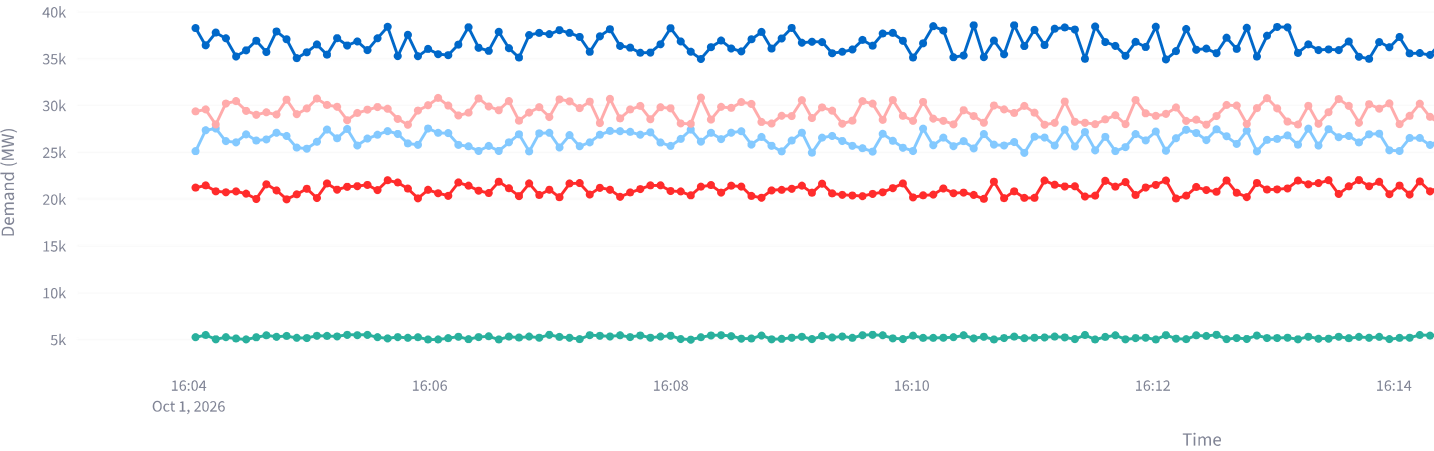
Real-Time Energy Demand, Generation and Electricity Price Analytics

Current Demand	Generation	DAM Price	Temperature	Avg Utilization
5,270 MW	2,147 MW	₹3.93/unit	36.4 °C	80.0%



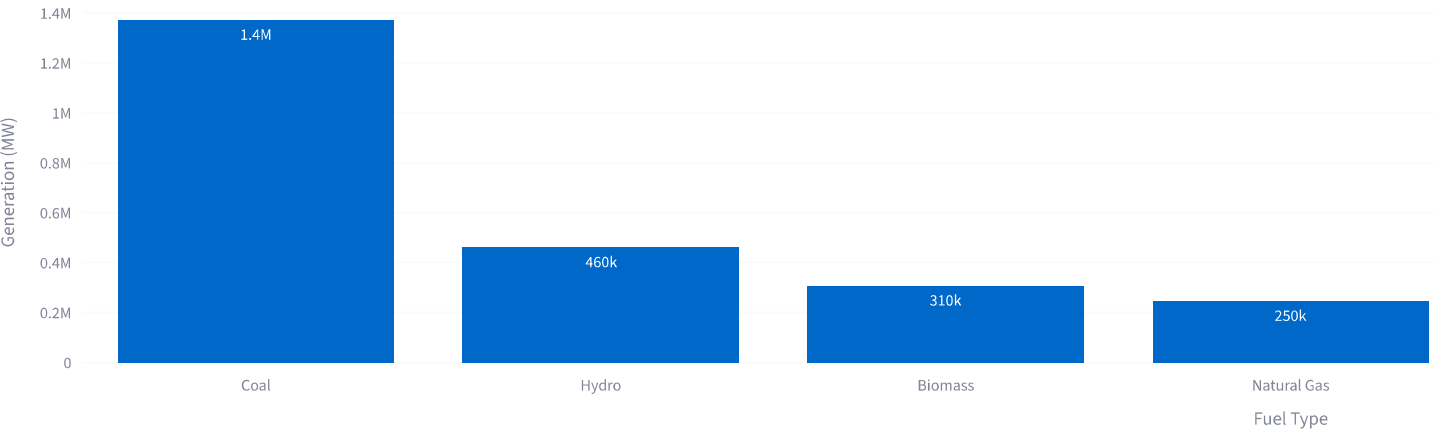
Electricity Demand Trend

Regional Electricity Demand



Power Generation by Fuel Type

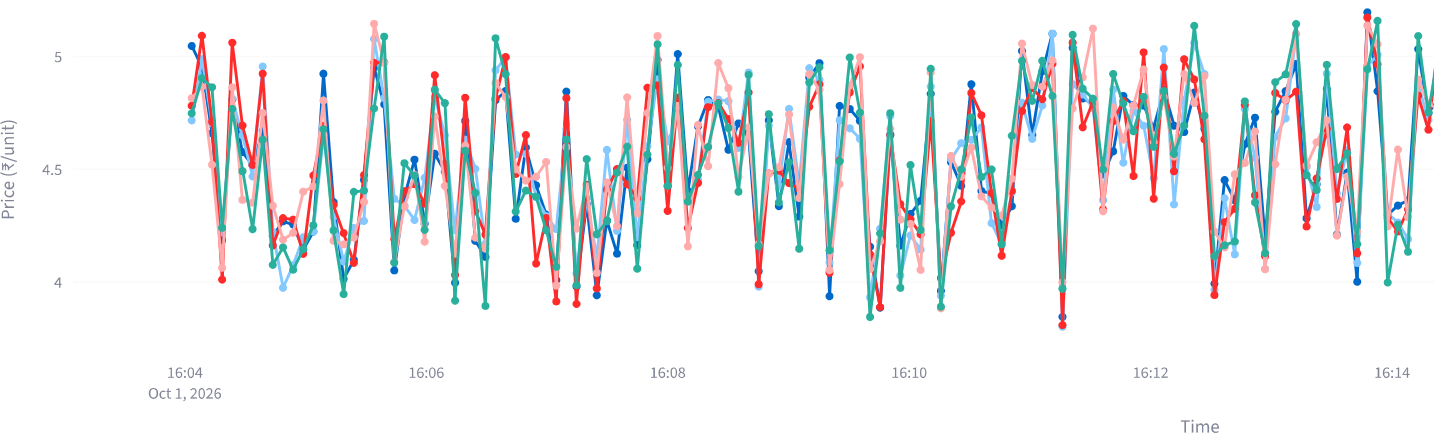
Generation by Fuel Type





Electricity Market Price

Day-Ahead Market Electricity Price



Temperature vs Electricity Demand

Average Temperature vs Average Demand



Business Insights

Energy Demand

Current observed demand: 5,270 MW

Regional demand monitoring helps identify changes in electricity load and supports grid planning.

Real-time demand analytics can help operators respond to changing consumption patterns.

Market and Generation

Current DAM price: ₹3.93 per unit

Current generation reading: 2,147 MW

Average generation utilization: 80.0%



Data Pipeline

Demand Records

1155

Generation Records

8085

Price Records

1155

Weather Records

1345



System Status

MongoDB: Connected

Kafka → MongoDB: Active

Streamlit Dashboard: Running

Last refresh: 2026-10-01 16:20:48 UTC

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